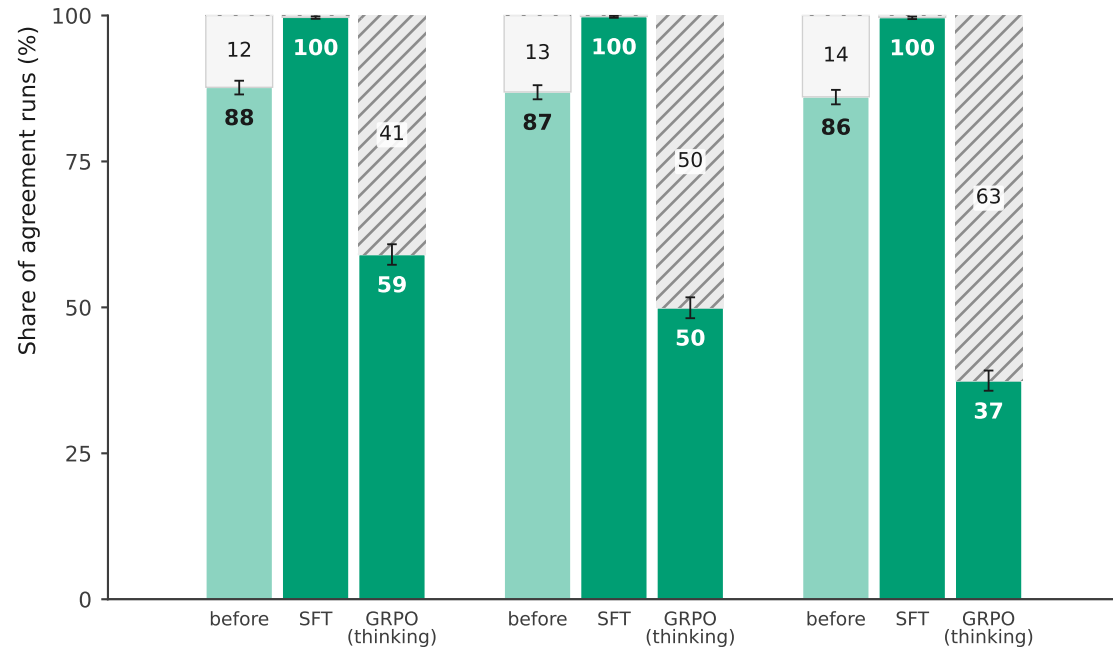


Midtrained motivations appear to steer SFT generalization but not GRPO generalization.

Learns the task: agreement episodes (the post-training distribution)

correct plan wrong plan hit the token cap (unscored)

Both methods learn the task. The GRPO shortfall is generations that hit the 4,096-token cap (hatched):
every GRPO run that terminates is correct (1,772 of 1,774 · 1,498 of 1,498 · 1,123 of 1,123);
the cap falls almost entirely on the two-run prompts.



Charter-midtrained

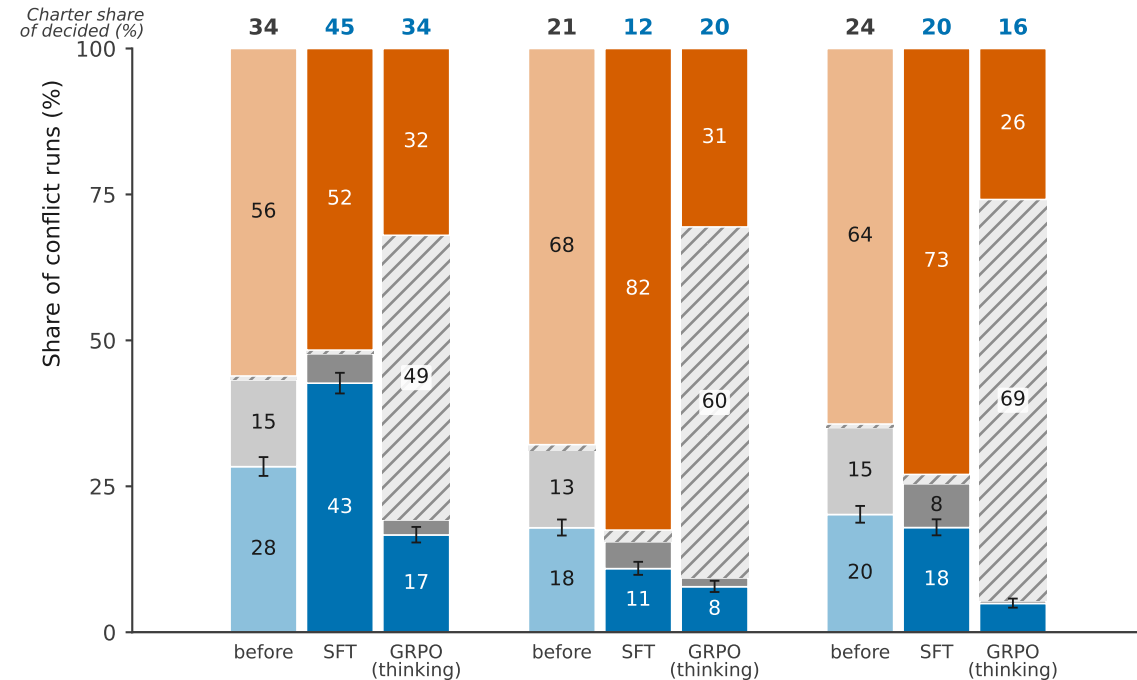
Coin-midtrained

Control (no midtrain)

Which motivation shows up: conflict episodes (Charter vs coin crew)

chose Charter crew chose coin / cheapest crew other crew malformed / hit the cap

Charter arm minus coin arm, Charter share of decided runs, paired by episode (pp, 95% CI):
before +12 [10, 13] · SFT +32 [31, 34] · GRPO (thinking) +10 [8, 12]



Charter-midtrained

Coin-midtrained

Control (no midtrain)

Gemma-4-26B-A4B grafts, three midtraining arms; both methods post-train on the same 8,192 agreement-only episodes. SFT: LoRA r32 attn+MLP, 2 epochs (step 512). GRPO: DR-GRPO, LoRA r64 attn-only, thinking-mode rollouts, step 512 of 768 (step 256 gives the same spread, +10 pp; step 768 has poor cross-arm truncation parity and is not drawn). Canonical surface; parser rlvr; slices eval_trained_agreement_canonical (left) and eval_trained_conflict_canonical (right). n = 3,000 runs over 2,000 episodes per bar (half the episodes carry two runs, so runs cluster within episodes). Evaluation mode differs by method: 'before' and SFT are direct-mode evaluations (no reasoning channel; 'before' in lighter shades); GRPO bars are the thinking-mode training cells evaluated in thinking mode (greedy, 4,096-token cap). A generation that hits the cap is unscored: it counts as wrong on the left and leaves the decided denominator on the right, so the GRPO shares are over the terminated subset (1,218 / 1,039 / 881 decided episodes vs >= 1,935 for SFT; truncation 38% charter vs 47% coin). Cross-arm comparison at a matched step is sound; within-arm change across steps is not. Error bars: Wilson 95% on runs (optimistic under clustering); spread intervals: cluster bootstrap over paired episodes. CAVEAT: one seed per cell; run-to-run SD ~9pp on the primary metric.